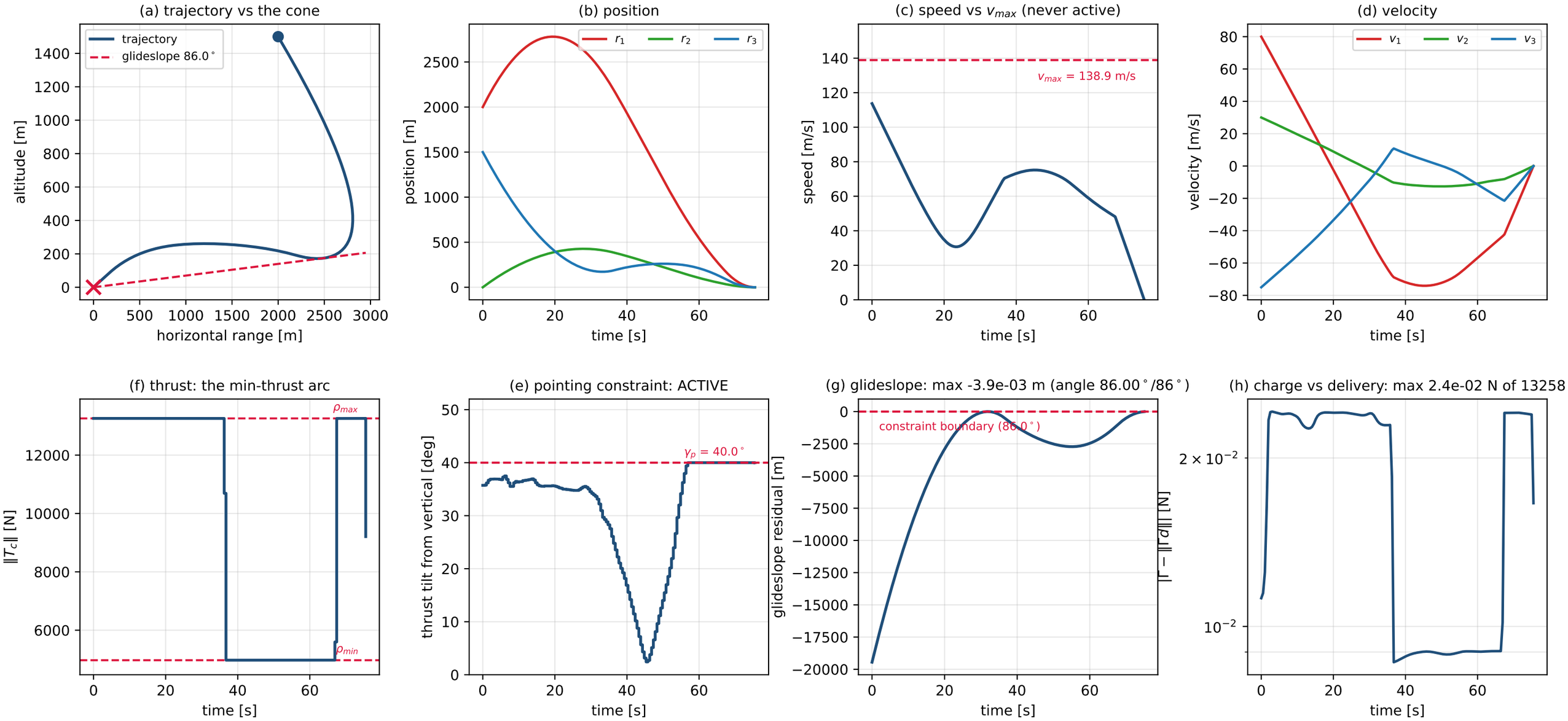
SCvx VERIFICATION, CONSTRAINED CASE - reproducing Fig. 21 of Malyuta et al. arXiv:2106.09125 (K = 151)
glideslope 86.0°, thrust tilt 40.0°, rotating frame ω = (3.5, 0, 2)×10^−3 deg/s; published t_f^* = 75.0 s
(a) trajectory vs the cone
trajectory
glideslope 86.0°
altitude [m]
horizontal range [m]
(b) position
r1
r2
r3
position [m]
time [s]
(c) speed vs v_max (never active)
v_max = 138.9 m/s
speed [m/s]
time [s]
(d) velocity
v1
v2
v3
velocity [m/s]
time [s]
(f) thrust: the min-thrust arc
ρ_max
ρ_min
‖T_c‖ [N]
time [s]
(e) pointing constraint: ACTIVE
γ_p = 40.0°
thrust tilt from vertical [deg]
time [s]
(g) glideslope: max -3.9e-03 m (angle 86.00°/86°)
constraint boundary (86.0°)
glideslope residual [m]
time [s]
(h) charge vs delivery: max 2.4e-02 N of 13258
2 × 10^−2
10^−2
‖T − ρ‖ [N]
time [s]